

AI-POWERED LOAN ANALYSIS SYSTEM

Product Requirements & Feature Specification Document

EXECUTIVE SUMMARY

Loan rejection is one of the most frustrating experiences for borrowers — and one of the most avoidable. The majority of rejected applications fail not because the borrower is ineligible, but due to incorrect bank selection, poorly structured files, or mismatches with automated underwriting systems.

This document specifies the requirements for building an AI-powered Loan Analysis System that transforms how rejected loan files are reviewed, scored, and re-routed — significantly increasing approval probability and processing speed.

PROBLEM → SOLUTION MAPPING

✗ PROBLEM	✓ AI SOLUTION
Incorrect bank selection	AI-driven bank/NBFC matching
Poor file presentation	Automated file quality scoring
Automated system mismatches	Profile-to-system compatibility check
Lack of CIBIL analysis	Real-time CIBIL evaluation
No structured review process	Structured rejection analysis workflow

EXPECTED OUTCOMES

85%	60%	3x
Approval Rate Improvement	Faster Processing Time	Better Bank Matching

CORE FEATURES TO BUILD

1 Rejection Reason Analyzer

Parses bank rejection notices and internal loan data to identify the primary reason(s) for rejection. Classifies into categories: CIBIL, income, documentation, bank mismatch, legal, technical.

2 Income & CIBIL Evaluator

Accurately calculates eligible loan amounts based on income type (salaried / self-employed / business), obligations, and credit score. Flags borderline cases and recommends improvement steps.

3 Bank & NBFC Matching Engine

Matches the borrower profile against a curated database of 50+ banks and NBFCs, ranked by approval probability. Considers LTV ratios, CIBIL cutoffs, income norms, and property type.

4 File Quality Scorer

Reviews the application file for completeness, document legibility, and formatting standards. Generates a File Quality Score (0–100) and a checklist of required corrections.

5 Re-submission Recommendation Report

Generates a structured PDF/Word report with: rejection analysis, corrective actions, recommended lenders, and revised loan eligibility — ready to share with the borrower.

AI SYSTEM PROMPTS TO IMPLEMENT

PROMPT 1 — Rejection Analysis Agent

You are an expert loan underwriter AI. Analyze the following rejected loan application and provide: 1. Primary rejection reason (1 sentence) 2. Secondary contributing factors (max 3 bullets) 3. CIBIL impact assessment (score range and effect) 4. Income gap analysis (required vs. actual) 5.

Documentation deficiencies (if any) Respond in structured JSON with keys: primary_reason, contributing_factors[], cibil_assessment, income_gap, doc_issues[].

PROMPT 2 — Bank Matching Agent

You are a loan placement specialist AI. Given the borrower profile below, identify the top 5 most suitable banks or NBFCs from the lender database. For each lender, provide: - Lender name and type (PSU / Private / NBFC) - Approval probability (%) - Key eligibility match points - Any risk flags Rank by approval probability (highest first). Respond in structured JSON.

PROMPT 3 — File Quality Audit Agent

You are a loan file audit specialist. Review the submitted loan application file and evaluate: 1. Document completeness (checklist of present vs. missing docs) 2. Data consistency (income, ITR, bank statements alignment) 3. File presentation score (0-100) 4. Critical corrections required before resubmission 5. Estimated resubmission readiness timeline Provide actionable, prioritized recommendations. Be specific and concise.

PROMPT 4 — Borrower Report Generator

You are a professional loan advisor AI. Generate a clear, empathetic borrower-facing report that explains: - Why the loan was rejected (in simple language, no jargon) - What the borrower can do to improve their profile - Which lenders are recommended for reapplication - Revised loan amount they may qualify for - Timeline to achieve approval readiness Tone: professional, encouraging, and actionable. Format as a structured letter.

REQUIRED DATA INPUTS

- Borrower KYC data (PAN, Aadhaar, age, employment type)
- CIBIL report or credit score (numeric + report PDF optional)
- Income documents: ITR, Form 16, salary slips, P&L statements

- Bank statements (6–12 months)
- Existing loan obligations / EMI schedule
- Property details (for home / LAP loans): type, location, valuation
- Rejection letter from previous lender (if available)
- Loan amount requested and tenure preference

RECOMMENDED TECH STACK

Layer	Recommended Technology
AI / LLM Engine	Claude Sonnet / GPT-4o via API — for analysis, matching, report generation
Backend	Node.js or Python (FastAPI) — API orchestration and business logic
Database	PostgreSQL — borrower profiles, lender matrix, application history
File Processing	PDF.js / PyMuPDF — document parsing and OCR
Report Generation	DOCX.js / ReportLab — structured output reports
Frontend	React.js + Tailwind — dashboard for loan officers and borrowers
Auth & Security	JWT + AES-256 encryption — PII data protection

SYSTEM WORKFLOW

1. Borrower / DSA uploads rejected file and loan details into the system
2. AI Rejection Analyzer extracts and classifies rejection reasons
3. CIBIL & Income Evaluator calculates revised eligibility
4. File Quality Scorer audits documents and generates correction checklist
5. Bank Matching Engine ranks top lenders by approval probability
6. System generates Structured Resubmission Report (PDF/DOCX)
7. Loan officer reviews and initiates re-application with recommended lender
8. Outcome is logged; system learns from approval/rejection feedback loop

Rejected loan files deserve structured review, not repeated rejection.

Smart analysis leads to better outcomes.